

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2022-2023

Comprehensive Regular
(EC-3 Examination)

Course No. : DSECLZG522
Course Title : Big Data Systems
Nature of Exam : Open Book
Weightage : 40%
Duration : 2.5 Hours
Date of Exam : 09/April/2023 (10:00 AM to 12:30 PM)

No. of Pages	= 3
No. of Questions	= 8

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

1. An eCommerce company is planning to launch a new application for online sale of items. During normal seasons of the year, this application deployed on a cluster of 4 servers will guarantee acceptable response to the customers. During festival seasons, the sale is expected to double. Which feature of cloud will be useful in minimizing the data center costs without compromising the response time. What are the recommended features of the components to be used in the design of this application for deployment on cloud?

(5 Marks)

2. What is the difference between a transformation and an action in Spark RDD, and how can you optimize the performance of your Spark RDD operations? What is meant by Lazy Evaluation?

(5 Marks)

3. An automobile manufacturing company is implementing a Cassandra based solution to automate their manufacturing process. Each machine used in the manufacturing process contains one or more sensors that captures data such as temperature, speed of a moving part, etc. This data is going to be stored in a Cassandra cluster that you must design and maintain. Once the application is up and running, the data it captures will be provided to analytics tool chains. The application will definitely be write heavy; tens of millions of events will be generated per second, and the company has an SLA requiring that all events be persisted across multiple nodes in the Cassandra cluster in less than 10ms. Suggest a suitable PRIMARY KEY (Primary and clustering columns) for the table given below:

```
CREATE TABLE sensor_data (  
    serial_number text,  
    date text,  
    snapshot_time timestamp,  
    facility_id int,  
    sensor_type text,  
    sensor_value text,  
);
```

(3 Marks)

4. A Bigdata analytics company is setting up an analytics platform for a new client. The requirement is to perform analytics on fixed size data of size 1 TB. After the initial days, the access to the data becomes infrequent and finally it can be moved to the Archive after a period of 360 days. The company decided to use AWS S3 for storing the data. The pricing for each of the S3 storage classes are given in the following table. The data access latency increases from Storage class 1 to 7. The company has accounted an amount of USD 2500 in the project for making use of the AWS S3 Storage.

No.	Storage Class	Cost in \$ / GB / Day	No. of days in storage Class	Cost in USD
1	S3 Standard	0.023		
2	S3 Intelligent - Tiering	0.023		
3	Standard - Infrequent Access	0.0125		
4	One Zone - Infrequent Access	0.01		
5	Glacier Instant Retrieval	0.004		
6	Glacier Flexible Retrieval	0.0036		
7	Glacier Deep Archive	0.00099	360 days	Total < \$2500

Define a storage policy to retain and move the data from Storage Class 1 to Storage class 7 to meet the following conditions:

1. Cost for storage should not exceed the allocated amount of \$2500 over a period of 360 days.
2. Data should move from storage class 1 to 6 (low to high) over a period of 360 days.
3. Data should be retained in each of the storage class 1 to 6 for a minimum period of 10 days.
4. The data should not be retained in two consecutive storage classes for equal number of days.
5. The number of days on which the data occupy the storage class from 1 to 6 should be in increasing order.
6. The policy should clearly state the number of days (Column 4 in table) in which data is retained in each storage class.
7. You need NOT consider the cost mentioned for Glacier Deep Archive since the data moves into this class after 360 days and remains there forever.

Credit will be given to a policy optimized to retain the data in the lowest classes for highest duration so that fast access to the data is available for a long duration.

You need to fill Columns 4 and 5 in the table with the number of days calculated for each storage class and the cost involved in keeping the data in that storage class for that many days.

(6 Marks)

5. Carefully go through scala Code1 and scala Code2 give below and answer the questions given below them.

Spark Scala Code1

```
val lines = new Array[String](2)
lines(0) = "Hello world"
lines(1) = "How are you world"
val stringRDD=sc.parallelize(lines,1)
val wordsRDD = stringRDD.flatMap(x => x.split(" "))
val wordRDD = wordsRDD.collect()
wordRDD
```

Spark Scala Code2

```
val lines = new Array[String](2)
lines(0) = "Hello world"
lines(1) = "How are you world"
val stringRDD=sc.parallelize(lines,1)
val wordsRDD = stringRDD.flatMap(x => x.split(" "))
val filtRDD = wordsRDD.filter(x => x.startsWith("H"))
val wordRDD = wordsRDD.collect()
wordRDD
```

Question 1. - Will there be any difference in the execution time for Code1 and Code2 ?

Question 2. - Modify Code1 to run on 2 executors in parallel

Question 3. - Modify the Code2 to output only words having a length of 5

Question 4. - Modify Code2 to output the number of unique words in the list.

(6 Marks)

6. How does Spark handle data partitioning and data locality in RDD, and what are the implications for performance and scalability?

(5 Marks)

7. A worker node used in a Hadoop cluster has 128 GiB memory with 8 CPUs having 6 cores each (Total of 48 cores). We need to reserve 1 core each for the Operating system, HDFS DataNode and YARN NodeManager. 8 Gib memory is reserved for the Operating system and 2 Gib each for the DataNode and the NodeManager. For handling the task overheads, 4 Gib memory is to be reserved. One container with Application Master (AM) may get deployed on this node. The AM container will require 2 Gib memory and 1 Core. (Total of 4 cores reserved for OS, DN, NM and AM).

1. How many vCores will be available for deploying containers for (1) CPU intensive tasks and for (2) standard I/O bound tasks.

2. If you are planning to use this node to launch containers for running standard I/O bound application tasks, give an estimate of the number of containers you will be deploying on this node. Describe a scheme for distributing the available resources (vCores and Memory) among these containers for achieving maximum performance.

(5 Marks)

8. A table named **world** is created in Hive using the following command:
create table world (country string, population int, GDP float, GINI float)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ',';

Data is loaded into this table from a CSV file.

Write HiveQL queries to find the following:

1. List of countries in which the population is more than 10 million
2. The country with the highest GDP
3. The country with the lowest value for the GINI coefficient
4. Name of 5 countries with the highest population.

(5 Marks)
